

Machine Learning Portfolio Project

1. Project Task

Complete a focused Machine Learning investigation using an approved dataset. Your project should normally address one classification **or** regression problem using techniques taught in the module. The aim is to demonstrate a sound process, not to use the most complex model or obtain the highest possible score.

You must:

1. Define a clear problem and explain the dataset;
2. Prepare the data and design a valid evaluation;
3. Build and compare a small number of suitable models against each other;
4. Interpret the results, limitations and responsible-use implications.

2. What you will Submit

- **Code or notebook:** a complete, readable workflow that loads the data, prepares it, trains the models and produces the reported results. You should use an online repository to track and store all the work for this project.
- **Detailed report:** maximum 3,000 words covering the problem, data, methods, results, limitations, and conclusion.
- **README and environment file:** brief instructions for running the work, including required packages.
- **Demonstration video:** A 5 minute video (screen capture with voice over) demonstrating your project and explaining your main decisions, results and one important limitation.
- **Assistance declaration:** Identify any external code, collaboration, generative AI or coding-assistant use throughout the project.

Use an approved, appropriately licensed dataset. Do not submit confidential or uncontrolled personal data. The submitted workflow must run using the instructions provided.

3. How the Project will be Graded

Criterion	Marks	What markers will look for
Problem and data understanding	3	A focused question; suitable dataset, target and features; clear source, context and limitations.
Data preparation and methodology	4	Appropriate cleaning and preprocessing; sensible train/test or cross-validation design; avoidance of data leakage; reproducible code.
Modelling and comparison	4	A simple baseline and at least two suitable learned models; justified choices; fair comparison and proportionate tuning.
Evaluation and critical analysis	5	Suitable metrics; clear comparison of results; error or residual analysis; conclusions supported by evidence.
Responsible ML and limitations	2	Intended use, important risks, interpretability and realistic limitations considered in the context of the project.
Communication and understanding	2	Clear report and visuals; well-organised submission; concise demonstration showing understanding of the work.
Total	20	

4. Alignment with Course Content

Project activity	Related module content
Define the task and inspect the dataset	Introduction to ML; Python and the ML ecosystem; data understanding.
Prepare features and design the evaluation	Data preparation and feature engineering; train/test design; metrics and leakage prevention.
Build and compare models	Regression, k-nearest neighbours, decision trees, ensembles and other suitable methods taught in the module.
Analyse and explain results	Model evaluation, error analysis, interpretability, responsible AI and critical reflection.

You are not required to use every algorithm taught. Select methods that suit your problem and explain your choices. It is also recommended that you research appropriate models outside of those taught in CT4101 in order to broaden your overall knowledge.

5. Work Plan

The following milestone submission deadlines will apply. Failure to submit a deliverable on time will result in a reduction of 4 marks from your final result. For example, if you only submit the final portfolio without any of the other deliverables, the most marks you can receive is 8/20 since you will have lost 3 x 4 for failing to submit the previous deliverables.

Milestone	Timing	Deliverable	Purpose
1. Project proposal	End of Week 3	One-page proposal identifying the dataset, prediction task, target variable, intended evaluation metrics and key data risks.	Confirms that the project is focused, feasible and uses an appropriate dataset.
2. Data and baseline checkpoint	End of Week 7	Working notebook or code showing the data audit, preprocessing pipeline, evaluation design and results from a simple baseline model.	Checks that the workflow is reproducible and that the evaluation avoids data leakage.
3. Model comparison checkpoint	End of Week 10	Draft results table comparing the baseline with at least two suitable learned models, plus one example of error or residual analysis.	Provides an opportunity to improve the analysis before final submission.
4. Final portfolio	End of Week 12	Complete code or notebook, report, README and environment file, demonstration video, and assistance declaration.	Demonstrates the complete machine learning process and is graded using the 20-mark rubric.

Milestones 1–3 are short progress deliverables used for formative feedback and project approval, but they are very important to ensure you are making consistent progress. They are marked on a complete/incomplete basis and failure to complete them will result in mark deductions as described above. The final portfolio is graded using the criteria in Section 3.

6. Exact Milestone Submissions

Submit each milestone through the designated Canvas assignment as a single ZIP file unless Canvas states otherwise. Use the filename format **StudentID_MilestoneNumber.zip**.

Milestone 1: Project Proposal

Submit **one PDF** of no more than one page containing:

- the project title and a clear classification or regression question;
- the dataset name, source, licence and citation;
- the target variable and main input features;
- the approximate number of records and important data-quality issues;
- the proposed baseline, candidate models and evaluation metrics;
- one short paragraph on privacy, fairness, misuse or other responsible-ML risks;
- a link to the online project repository.

Milestone 2: Data and Baseline Checkpoint

Submit:

- a working notebook or source-code folder that loads the data and runs the work completed so far;
- a README with setup and run instructions;
- an environment or requirements file listing required packages;
- a brief data audit covering dataset shape, variable types, missing values, duplicates, target distribution and any obvious outliers;
- the preprocessing pipeline and explanation of the train/test split or cross-validation design;
- results for a simple baseline model using the proposed metrics;
- the repository link and a short assistance declaration.

Milestone 3: Model Comparison Checkpoint

Submit:

- updated runnable code or notebook;
- a results table comparing the baseline with at least two suitable learned models under the same evaluation protocol;
- the main model settings and a brief explanation of any tuning performed;
- at least one relevant visualisation, such as a confusion matrix, ROC or precision–recall curve, residual plot or feature-importance plot;
- a short error or residual analysis identifying where a model performs poorly;
- a draft conclusion stating the current best model, evidence, limitations and next steps;
- the updated repository link and assistance declaration.

Milestone 4: Final Portfolio

Submit:

- the final code or notebook, with outputs generated from the submitted workflow;
- the final report of no more than 3,000 words;
- a README and environment or requirements file sufficient to reproduce the work;
- any small permitted data files, or clear instructions for obtaining the dataset where redistribution is not allowed;
- the final results, visualisations, error analysis, responsible-ML discussion and limitations;
- a five-minute demonstration video submitted in the format specified on Canvas;
- the final repository link;
- a complete assistance declaration covering external code, collaboration, generative AI and coding-assistant use.

Before submitting each milestone, confirm that files open correctly, links are accessible to the teaching team and the supplied instructions run the work without undocumented steps.

7. Important Information

- The work must be **your own** and you must be able to explain all submitted code, results and decisions. Suspicious cases may lead to student interviews and could result with being referred to the School of Computer Science Academic Integrity Advisor for disciplinary action.
- Do not tune models using the final test set.
- Cite the dataset and any external code or sources.
- Declare any use of generative AI and coding assistants, and verify all assisted work.

8. Resources and Setup Guides

The following resources can help you find a suitable dataset and create a reproducible project. Always check the dataset licence, documentation, provenance and ethical implications before requesting approval.

Finding an Appropriate Dataset

- **UCI Machine Learning Repository:** well-documented datasets for classification and regression, often with clear feature and target descriptions.
- **OpenML:** searchable datasets with machine-readable metadata, task types and links to previous experiments.
- **Kaggle Datasets:** a large collection of community and organisation-published datasets. Check the original source, licence and quality rather than relying only on the Kaggle description.
- **data.gov.ie:** the Irish Government open-data portal, useful for public-sector, transport, environment, health and regional datasets.
- **Central Statistics Office Ireland:** official Irish statistical datasets. Confirm that the selected table supports a meaningful prediction task rather than only descriptive analysis.
- **EU Data Portal and other official open-data portals:** useful for public datasets with documented provenance and reuse conditions.

A suitable dataset should have a clear prediction target, enough usable records and features for model comparison, manageable size, documented meaning, and a licence that permits academic use. Avoid datasets containing uncontrolled personal, confidential or highly sensitive information.

Repository Setup

- Use the official **GitHub Docs: Creating and managing repositories** guide to create a repository.
- Add a **README** explaining the project question, dataset source, folder structure, setup steps and exact commands required to run the work.
- Use a Python **.gitignore**, following the official **GitHub Docs: Ignoring files** guide. Do not commit virtual environments, passwords, API keys, temporary files or large generated outputs.
- Commit **regularly** using short, meaningful messages. **The history should show steady development rather than one final upload.**
- Do not place data in the repository unless its licence permits redistribution. Otherwise, provide download and preparation instructions.

Python Environment

- Create a separate environment for the project using either Python **venv** or **conda**. The official **scikit-learn installation guide** provides current commands for Windows, macOS and Linux.
- Install only the packages required by the project and record their versions in **requirements.txt** or **environment.yml**.
- Test the recorded environment by recreating it and running the full workflow from a clean session before each milestone.
- Record the Python version and any system-specific requirements in the README.

Notebooks and Code

- Jupyter Notebook or JupyterLab may be used locally; a hosted notebook such as Google Colab may also be used if the work can be reproduced from the submitted files and environment information.
- Organise notebooks in a clear sequence: project setup, data loading, audit, preprocessing, modelling, evaluation and conclusions.
- Use descriptive headings and short explanations, but keep code cells focused and remove abandoned experiments.
- Restart the kernel and run all cells from top to bottom before submission. The notebook must not depend on hidden variables or an undocumented execution order.
- Where the workflow becomes lengthy, move reusable functions into Python files and import them into the notebook.

Suggested Project Structure

Item	Purpose
README.md	Project overview, dataset source, setup and run instructions.
requirements.txt or environment.yml	Package names and versions required to reproduce the work.
notebooks/	Numbered notebooks for exploration, modelling and final analysis.
src/	Reusable Python functions or scripts, where needed.
data/README.md	Dataset source, licence, download and preparation instructions; avoid committing restricted or large data.
reports/	Final report and any figures required for submission.
.gitignore	Files and folders that must not be tracked.

Use the latest versions of official documentation. Search for the exact resource titles above rather than following unverified installation instructions from third-party sites.